

10 D.C. circuits

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Kirchhoff's first law conserves charge at a junction; the second conserves energy around a loop. In series, current is shared and potential differences add; in parallel, potential difference is shared and currents add. A source with internal resistance r has terminal p.d. $V = \varepsilon - Ir$ during discharge.

Worked example: follow the reasoning

A 12 V source with $r = 1\ \Omega$ across a $5\ \Omega$ load gives $I = 12/(5+1) = 2\text{ A}$ and terminal p.d. = 10 V.

Quick check · multiple choice

A How is terminal p.d. of a discharging cell related to e.m.f.?

[1]

A $V = \varepsilon + Ir$

B $V = \varepsilon - Ir$

C $V = Ir - \varepsilon$

D $V = \varepsilon / I$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A cell has $\varepsilon = 9.0\text{ V}$ and internal resistance $0.50\ \Omega$; external resistance is $4.0\ \Omega$. Find current and terminal p.d. [3]

Final answer: _____

Answers and explanation

Quick check: B. $V = \mathcal{E} - Ir$

Question 1 [3 marks]

$I = \mathcal{E}/(R+r) = 9/4.5 = 2.0 \text{ A}$ [2]; $V = IR = 8.0 \text{ V}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.